Preliminary Literature Map



ACE

CYBER R&D Program

Conversational AI Interface for
Analyst-Driven Threat Investigation

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1.0 Research Focus

1.1 Research Question

How can a Conversational AI interface specifically integrated with Elasticsearch be designed to enable SOC analysts to query log data using natural language for common adversarial threat-hunting tasks (e.g., failed logins, suspicious IPs)?

1.2 List of Keywords

Cybersecurity | Anomaly detection | Conversational AI | Natural language Querying | SOC log analysis | Elasticsearch threat hunting | Log data analysis automation | AI-assisted intrusion detection | Human-AI collaboration in SOCs | Cybersecurity decision support systems | SOC analyst efficiency | False positives in intrusion detection | AI for adversarial detection | Cognitive workload SOC analysts | AI-augmented security operations | Chatbot cybersecurity applications | Explainable AI in threat detection | Automated log query systems | AI usability cybersecurity | LLMs for cyber defense

2.0 Preliminary Themes & Gaps

2.1 Emerging Themes

- There is strong evidence from multiple studies that AI is effective for detecting cyber threats and anomalies. Models such as RNN, LSTM, and CVAE significantly improve accuracy and prediction.
- Research shows analysts often distrust AI alerts due to cognitive biases, high false positives, and poor explainability. Studies on XAI, hybrid human-AI workflows, and user-centric design stress transparency, trust, and usability. Human factors are increasingly recognized as critical for AI adoption in operational SOCs.
- Practical integration of AI into existing SOC tools and workflows remains limited.

2.2 Potential Research Gap

- Most AI research focuses on anomaly detection, model accuracy, or autonomous detection, but there is a lack of systems allowing analysts to query log data via natural language.
- Human–AI collaboration is often theoretical; interactive tools that improve efficiency and analyst satisfaction are scarce.
- Many papers measure model accuracy and speed but neglect analyst-centric metrics, including trust, satisfaction, error reduction, and interpretability in decision-making.

3.0 Supporting Literature Table

3.1 Source #1

Citation: W. Villegas-Ch, A. Jaramillo-Alcázar, A. Maldonado Navarro, and A. Mera-Navarrete, "Integrating Explainable Artificial Intelligence in Anomaly Detection for Threat Management in E-Commerce Platforms," IEEE Access, vol. 13, pp. 29830–29846, 2025, doi: 10.1109/ACCESS.2025.3541979.

Research Type: Applied Research

Key Findings: Achieved 95% precision with explanation generation taking less than 500ms. Reduced analyst validation time by 33%. Increased confidence by 20%. Outperformed traditional clustering methods.

Relevance: The study by W. Villegas-Ch highlights the importance of combining advanced AI with human-centered workflows in cybersecurity, demonstrating how explainable models (utilizing SHAP, LIME, and Grad-CAM) can enhance analyst trust and efficiency. This research connects directly to the present study, which also aims to support SOC analysts through a conversational AI interface that simplifies querying Elasticsearch logs. Both emphasize speed, accuracy, and usability as critical factors for operational environments.

Limitations: The study's focus on e-commerce environments limits its direct applicability to SOC operations. Although it addresses interpretability, it lacks mechanisms for interactive querying or domain-specific integration with platforms like Elasticsearch. Furthermore, its evaluation prioritizes precision and explanation speed but does not fully measure analyst satisfaction or workflow usability. The present research fills these gaps by

targeting SOC-specific contexts, prioritizing natural language interaction, and explicitly evaluating analyst performance and satisfaction.

3.2 Source #2

Citation: S. Samtani, M. Kantarcioglu, and H. Chen, "Trailblazing the Artificial Intelligence for Cybersecurity Discipline: A Multi-Disciplinary Research Roadmap," ACM Trans. Manage. Inf. Syst., vol. 11, no. 4, pp. 1–19, Dec. 2020, doi: 10.1145/3430360.

Research Type: Theoretical Research

Key Findings: AI has significant potential in cybersecurity for tasks such as asset prioritization, vulnerability management, and threat detection.

Relevance: This article by S. Samtani establishes the broader foundation for why AI is needed in cybersecurity and why practical, cross-disciplinary applications are crucial. The present research contributes directly to this roadmap by creating a conversational AI system for SOC analysts, bridging AI techniques with operational cybersecurity workflows.

Limitations: The article is conceptual and broad, offering a high-level roadmap without focusing on specific implementations like SOC log querying or natural language interfaces. It emphasizes long-term research directions but does not provide concrete benchmarks (e.g., latency, detection accuracy, user satisfaction). It lacks a practical proof of concept, leaving open the question of how AI tools can be effectively integrated into day-to-day analyst workflows.

3.3 Source #3

Citation: N. Roch, "Building Trust Bridges: The Dynamics of Autonomy and Transparency in Expert-AI Collaboration for Cybersecurity," in Companion Proceedings of the 30th International Conference on Intelligent User Interfaces, in IUI '25 Companion. New York, NY, USA: Association for Computing Machinery, Mar. 2025, pp. 202–204. doi: 10.1145/3708557.3716152.

Research Type: Observational and Applied Research

Key Findings: AI can assist with routine cybersecurity tasks, but cannot replace expert judgment. Trust and transparency are essential for effective human–AI collaboration. Explainable AI (XAI) tools may improve analyst trust and adoption of AI in operations.

Future evaluations will explore how varying levels of AI autonomy affect expert acceptance.

Relevance: This work by N. Roch highlights the importance of trust in AI systems for SOC analysts, which directly connects to the present research. Both pieces of work similarly aim to augment analysts rather than replace them, improving usability and trust through natural language interaction. Both emphasize building systems that analysts can validate and confidently rely on.

Limitations: Their research focuses primarily on trust, transparency, and adoption rather than operational performance metrics like query latency or detection accuracy. It does not address log querying or SOC-specific integration with tools like Elasticsearch, which is central to the present research. The XAI approach improves interpretability but does not directly provide interactive natural language mechanisms for analysts to engage with data.

3.4 Source #4

Citation: S. Nsvsc Ramesh et al., "Leveraging Cyberattack News Tweets for Advanced Threat Detection and Classification Using Ensemble of Deep Learning Models With Wolverine Optimization Algorithm," IEEE Access, vol. 13, pp. 48343–48358, 2025, doi: 10.1109/ACCESS.2025.3550378.

Research Type: Applied Research

Key Findings: The ETIC-EDLMOA model (ensemble of RNN, LSTM, CVAE, optimized with WoOA) achieved 98.51% accuracy in detecting and mitigating attacks. Data preprocessing and feature extraction (Word2vec) improved classification performance. Demonstrates the potential of combining deep learning with metaheuristic optimization for CTI.

Relevance: This study by S. Nsvsc Ramesh demonstrates the effectiveness of advanced AI methods in analyzing large, complex cybersecurity datasets, which parallels the present project's aim of improving SOC workflows through AI. While their focus is on detection accuracy, the present project's research focuses on accessibility and usability, which enables analysts to query log data in natural language. Both works, however, align in addressing the broader challenge of augmenting analyst capabilities with AI-driven tools for faster and more accurate threat detection.

Limitations: The study focuses heavily on technical model accuracy without considering human usability, transparency, or analyst interaction. It operates in a simulation

environment, which may not fully capture SOC operational realities. The model is optimized for classification tasks, but it does not address query efficiency, user satisfaction, or SOC-specific integration with platforms like Elasticsearch.

3.5 Source #5

Citation: A. Horváth et al., "Anomaly Detection Algorithms for Real-Time Log Data Analysis at Scale," IEEE Access, vol. 13, pp. 136288–136311, 2025, doi: 10.1109/ACCESS.2025.3594469.

Research Type: Applied Research

Key Findings: Advanced methods like CAELAD achieve high accuracy, but simpler models like SCLAD can be equally effective depending on context. The proposed algorithms improved F1-scores by 6–10% over existing methods. Anomaly detection in real-world IT and cloud environments requires balancing complexity and simplicity to match system requirements. Practical applicability depends not only on accuracy but also on adaptability to different environments.

Relevance: This work by A. Horváth reinforces the importance of log-based anomaly detection in real-world operational environments, which aligns directly with this research's focus on SOC logs. It shows that log data can be effectively analyzed using AI, validating the core of this research. Importantly, the emphasis on balancing complexity and simplicity parallels the approach of this research. Both directions highlight that operational value requires more than just sophisticated models; it requires integration into workflows that analysts can actually use.

Limitations: The study focuses narrowly on algorithm performance (F1-score, accuracy) but does not address human interaction, trust, or usability. It does not provide mechanisms for natural language querying or analyst-facing interfaces. The evaluation is limited to detection tasks and does not consider broader SOC metrics, such as query latency, user satisfaction, or analyst performance improvement. While the paper shows practical improvement, it lacks discussion on integration with SOC platforms (for example, Elasticsearch).

3.6 Source #6

Citation: R. A. Hagen, L. Øverlier, and K. Helkala, "Human Factors in AI-Driven Cybersecurity: Cognitive Biases and Trust Issues," Digital Threats, p. 3759260, Aug. 2025, doi: 10.1145/3759260.

Research Type: Observational Research

Key Findings: Analysts experience significant cognitive biases when interacting with AI tools: automation bias (47%) and confirmation bias (37%).
65% of analysts distrust AI alerts, citing false positives and poor explainability.
79% prefer hybrid human–AI models over full automation.
Strategies like explainable AI, bias-awareness training, and adaptive trust calibration can improve adoption and effectiveness.

Relevance: This study by R. A. Hagen highlights the human trust and usability challenges that directly affect the present research's goal of creating a conversational AI for SOC log querying. The findings support the present research's emphasis on building analyst-friendly, interactive tools rather than fully automated systems. By prioritizing natural language interfaces and usability metrics (latency, accuracy, satisfaction), the present research aligns with the call for user-centric AI that respects analyst workflows and cognitive realities.

Limitations: The study is observational, focusing on interviews and comparative analysis rather than testing new systems or prototypes. It does not propose or evaluate a concrete technical solution, unlike the present research's applied experimental approach. While it identifies trust issues and cognitive biases, it does not address practical mechanisms (e.g., natural language querying, SOC integration with Elasticsearch) to overcome them. The evaluation is limited to perceptions and attitudes, not quantitative performance metrics like latency or detection accuracy that the present project measures.

3.7 Source #7

Citation: M. Gupta, C. Akiri, K. Aryal, E. Parker, and L. Praharaj, "From ChatGPT to ThreatGPT: Impact of Generative AI in Cybersecurity and Privacy," IEEE Access, vol. 11, pp. 80218–80245, 2023, doi: 10.1109/ACCESS.2023.3300381.

Research Type: Theoretical Research

Key Findings: Generative AI can be exploited through jailbreaks, prompt injection, and reverse psychology attacks. Offensively, GenAI can be leveraged for phishing, social engineering, malware creation, and automated hacking. Defensively, GenAI has potential for threat intelligence, automated defense, secure code generation, and incident response. Raises important social, legal, and ethical concerns about safety, trust, and

misuse. Identifies open challenges and future research directions to make GenAI trustworthy and secure.

Relevance: This work highlights both the opportunities and risks associated with deploying conversational AI in cybersecurity. The defensive applications discussed (threat intelligence, log analysis, automated reporting) reinforce the value of conversational AI for SOCs, while the highlighted risks provide critical guidance on risk mitigation strategies in the present project's implementation.

Limitations: The study is primarily conceptual and review-based, without empirical testing of defenses or performance evaluation. It focuses broadly on GenAI as a cybersecurity threat and tool, but does not address log querying, Elasticsearch integration, or SOC analyst workflows. While it identifies vulnerabilities in GenAI, it does not propose concrete, domain-specific safeguards for SOC environments.

3.8 Source #8

Citation: F. Guan, H. Ye, J. Zeng, J. Liu, C. Li, and W. Wen, "Research on AI-Driven Intelligent Perception of Cybersecurity Threats and Dynamic Defense Mechanisms," in Proceedings of the 2025 2nd International Conference on Communication Security and Information Processing, in CSIP '25. New York, NY, USA: Association for Computing Machinery, July 2025, pp. 39–47. doi: 10.1145/3744668.3744675.

Research Type: Applied Research

Key Findings: The proposed architecture achieved 95% detection accuracy and reduced average response delay to under 5 seconds, a 55% improvement over conventional systems.Traffic prediction error decreased by 21% compared to ARIMA benchmarks. Stress testing (10Gbps DDoS attacks) revealed performance degradation (accuracy drop of 18%), highlighting limits under extreme conditions.
Demonstrates that AI-driven adaptive frameworks can improve both speed and effectiveness in detecting sophisticated cyberattacks.

Relevance: This study is relevant because it demonstrates the impact of AI on reducing response times and improving detection accuracy in cybersecurity operations, which aligns with the present project's goal of accelerating SOC analyst workflows. The focus on timely detection and handling of complex attacks parallels the present research's metrics for latency (below 5 seconds) and accuracy (above 80%) in querying log data. Both approaches aim to enhance human-AI collaboration for operational effectiveness.

Limitations: The study by F. Guan focuses on automated threat detection rather than interactive human-AI interfaces, so it does not address natural language querying or analyst usability. While it reports performance metrics under simulation, it does not evaluate analyst satisfaction, trust, or workflow integration. The architecture's complexity and stress test vulnerabilities indicate potential challenges for real-world SOC deployment, which the present project's PoC addresses with a scoped implementation. It does not integrate with specific platforms like Elasticsearch, limiting direct applicability to SOC log querying.

3.9 Source #9

Citation: K. Dhanushkodi and S. Thejas, "AI Enabled Threat Detection: Leveraging Artificial Intelligence for Advanced Security and Cyber Threat Mitigation," IEEE Access, vol. 12, pp. 173127–173136, 2024, doi: 10.1109/ACCESS.2024.3493957.

Research Type: Theoretical Research

Key Findings: AI, particularly machine learning and deep learning, significantly improves threat detection, including network breaches, adversarial attacks, and zero-day vulnerabilities. Explainability and resilience are critical for trustworthy AI in security applications. Emerging techniques like transformers, federated learning, and blockchain integration enhance real-time and robust threat detection. Challenges remain in large-scale data management, real-time processing, and privacy/security considerations.

Relevance: This review provides context for the importance of AI in operational cybersecurity and highlights the need for systems that are both effective and trustworthy, which aligns with the present research's goal of creating a conversational AI interface for SOC analysts. The present project directly addresses these principles by prioritizing usability, latency, and analyst trust while interacting with large-scale log data. The discussion of real-time processing challenges also reinforces the relevance of the present project's latency performance metrics (below 5 seconds).

Limitations: The paper is a review, so it does not propose or evaluate a specific system or interface. It focuses broadly on AI applications across multiple sectors and not specifically on SOC log querying or Elasticsearch integration. While it emphasizes explainability and resilience, it does not provide practical methods for natural language interaction or analyst workflow integration. Metrics for human analyst satisfaction, query efficiency, or usability are not addressed.

3.10 Source #10

Citation: N. Alsharabi, A. Bhardwaj, T. Sarheed Alshammari, S. Alotaibi, D. Alshammari, and A. Jadi, "Threat Hunting the Shadows: Detecting Adversary Lateral Movement With Elasticsearch," IEEE Access, vol. 13, pp. 62341–62352, 2025, doi: 10.1109/ACCESS.2025.3556184.

Research Type: Experimental Research

Key Findings: Identifies effective methods for detecting lateral movement within compromised networks. Highlights the importance of credential access monitoring as a critical point for detecting adversary activity. Demonstrates that SIEM systems integrated with Elasticsearch and advanced query languages (KQL, Lucene) can proactively identify indicators of compromise (IoCs). Simulation-based evaluation provides practical insights into designing targeted search queries for threat detection.

Relevance: This work is directly relevant because it focuses on log querying and threat detection in a SOC-like environment using Elasticsearch, which mirrors the present project's core functionality. It demonstrates how structured queries can uncover adversary TTPs, reinforcing the need for a conversational interface that allows analysts to perform these tasks efficiently. The research supports the present project's goal of improving query accessibility, latency, and accuracy for SOC analysts.

Limitations: The study relies on custom SIEM implementations and advanced queries, but does not explore natural language interaction for analysts.
Evaluation is based on simulated datasets, so real-world usability and analyst workflow integration are not tested. Focused primarily on lateral movement detection, limiting generalizability to other common SOC queries (e.g., failed logins, suspicious IPs).
Does not measure human factors such as analyst satisfaction, trust, or efficiency.

3.11 Source #11

Citation: A. Bhardwaj, L. Sapra, and S. Rahman, "Elasticsearch-Based Threat Hunting to Detect Privilege Escalation Using Registry Modification and Process Injection Attacks," Future Internet, vol. 17, no. 9, p. 394, Sept. 2025, doi: 10.3390/fi17090394.

Research Type: Applied Research

Key Findings: ASEP Registry Key Modifications Are Detectable: Unauthorized modifications to AutoStart Extension Points during user logon are a reliable indicator of persistence attempts.
DLL Injections in Temporary Directories Are Critical Indicators: Malicious DLLs executed from transient directories can be tracked via Sysmon Event IDs.
Event Correlation Enables Rapid Detection: Correlating Sysmon events (registry creation/modification, process creation, DLL loading) allows identification of attack chains in near real-time.
SIEM Integration Improves Mitigation: Using Elasticsearch for structured event collection and correlation enhances the ability to disrupt adversarial workflows quickly.

Relevance: Demonstrates practical methods for threat-hunting analysts to identify persistence and privilege escalation in Elasticsearch enterprise environments.
Highlights the importance of monitoring registry and filesystem activities to detect attacks before they fully compromise systems.
Provides a foundation for natural language or AI-assisted querying of SIEM logs by showing which event types are most informative.

Limitations: Focuses specifically on Windows registry and filesystem events; may not correlate to Linux/macOS.
It does not deal with AI integration; instead, it serves as a good example of manual Elasticsearch querying.

3.12 Source #12

Citation: Q. Zheng, Y. Tang, Y. Liu, W. Liu, and Y. Huang, "UX Research on Conversational Human-AI Interaction: A Literature Review of the ACM Digital Library," in CHI Conference on Human Factors in Computing Systems, New Orleans, LA, USA: ACM, Apr. 2022, pp. 1–24. doi: 10.1145/3491102.3501855.

Research Type: Systematic Literature Review

Key Findings:
The review identified four primary effects of conversational agents in polyadic contexts: communication, engagement, connection, and relationship maintenance. A consistent theme was the importance of trust, transparency, and boundary awareness, with users expecting clarity about agent roles, data handling, and transitions between different interaction modes. The study also catalogued evaluation methods used in the literature,

such as log analysis, surveys, and interviews, while highlighting the lack of standardized usability and trust assessments.

Relevance: Provides a human–computer interaction (HCI) framing for conversational interfaces. It highlights that success should not be measured solely by technical performance but also by usability and trust, which are crucial for adoption in high-stakes environments like SOCs. The evaluation approaches discussed by Zheng et al. also offer practical methods for assessing analyst experiences
alongside performance metrics such as query latency and accuracy.

Limitations: It is domain-general and does not address SOC workflows, Elasticsearch integration, or adversarial threat-hunting tasks.

3.0 Change Log

Ver.	Date	Description of Change	Author
v1.1	2025-09-17	Filled out document (first time)	Koi McFarland
v1.2	2025-09-27	Found two extra sources	Koi McFarland
v1.3	2025-11-19	Updated Formatting	Koi McFarland

4.0 Mentor Review / Approval

Reviewer	Role	Date	Notes
Student Peer	PI	2025-12-17	Final Deliverable
Faculty Advisor			
DoD/ACE Mentor			